

Assignment-3 For Loop in Python

1. Write a Python program to print all odd and even numbers from 1 to 20.
2. Write a Python program to generate all multiples of 12.
3. Write a Python program to generate a table of a number provided by the user.
4. Write a Python program to check if a number provided by the user is prime or not.
5. Write a Python program to calculate the sum of numbers between a starting and ending point provided by the user.
6. Write a Python program to calculate the product of numbers between a starting and ending point provided by the user.
7. Write a Python program to generate the Fibonacci sequence up to a specified number of terms.
8. Write a Python program to calculate the factorial of a number provided by the user.
9. Write a Python program to find the greatest character from the string "python".
10. Write a Python program to display all letters except 'm' and 'i' from the string "Dreamer infotech".
11. Write a Python program to print alternate characters from a given string.
12. Write a Python program to reverse a string entered by the user.
13. Write a Python program to count the total number of characters in a string entered by the user.

14. Write a Python program to check whether a string entered by the user is a palindrome.
15. Write a Python program that allows the user to search for a character within a given string.
16. Write a Python program to filter out all vowels and consonants from a string entered by the user.
17. Write a Python program to filter out duplicate characters from a string entered by the user.
18. Write a Python program to display all possible pairs of 3.
Example: 1:3, 2:3, 3:3, 2:1, 2:2, 2:3, 3:1, 3:2, 3:3
19. Write a Python program to generate the pattern of the letter H.
20. Write a Python program to find duplicate letters between two strings.
Example: In "virat" and "rohit", the duplicate letter is "r".
21. Write a Python program to display the squares of numbers from 1 to 10.
22. Given a string `text = "python"`, calculate the **sum of the indices** of its characters **without using** the `range()` or `len()` functions.
23. **Given:** `text = "python programming"`
Goal: Count how many **vowels** are in the string.
Constraint: Do **not** use indexing (`text[i]`) or slicing (`text[:]`).
24. **Given:** `text = "programming"`
Goal: Print all characters that repeat in the string.
25. **Given :** 01275623
Write a Python program to find and print the **greatest character** in the string.

26. **Given:** `text = "knowyourself"`

Goal: Find and print the **first character** that repeats.

27. Give : `text="if you think you can not do, you can not show think wisely"`

Goal: Print the alternate words

Constraint: Do **not** use space between words more than once .

28. **Given:** `text = "knowyourself"`

Goal: Find and print the **alternate characters**.

29. Take two numbers from the user: start and end. Print a string labeling each number in that range as Odd or Even.

Output_format : 3:Odd 4:Even 5:Odd 6:Even 7:Odd 8:Even 9:Odd

30. Find the sum of string "198765456412".

31. Count how many digits in the string are greater than 5 from `text = "1234567890"`.

32. Task: Replace Character in String

Write a program that takes a string input from the user, then asks for a character to replace and the character to replace it with. The program should output the modified string where all occurrences of the specified character are replaced by the replacement character.

33. Replace Spaces with Underscores Replace all spaces in a string with underscores (`_`)

34. Remove Duplicate characters from the string given by the user then print the final output.

35. Take string from user and Replace every vowel in the string with an asterisk `*`.

36. Count only words not spaces.

Entered a string: Hello coders from Success24

Number of words: 4

37.**Task:** Count how many uppercase and lowercase letters are in a string.

38.**Task:** Print all characters from the string that are at odd indices.

39.**Task:** Create a new string by removing all spaces from the input string.

```
Enter a string:  how are you all
String without spaces: howareyouall
```

40.Count how many digits in a string entered by the user.

```
text="syntax_error2806 hai "
```

41.Count only alphabets (both uppercase and lowercase).

```
Enter a string:  this123 i am
Number of letters: 7
```

42.You are managing a simple banking system that tracks the balance at the end of each day over 10 days. Each day, the balance increases by 100 units starting from 100 on day 1, 200 on day 2, and so on. You want to print the current day's balance along with the previous day's balance. For day 1, the previous day's balance is 0.

Sample output:

Day 1: Balance = 100, Previous Day Balance = 0

Day 2: Balance = 200, Previous Day Balance = 100

Day 3: Balance = 300, Previous Day Balance = 200

...

Day 10: Balance = 1000, Previous Day Balance = 900